

EEE3096/5S Embedded Systems II

Lecture 03 : Development & Execution Environments

Take-home Activity: Thinking of Development & Execution Environment for an embedded system

Activity:

Imagine that you are a chief engineer in an embedded systems development company. The company happens to develop a variety of corporate gifts, but the company is proud of producing gifts that are more useful to the receiver, not the usual kind of marketing stuff that can be somewhat tedious, take-up precious suitcase space ... and are sometimes 'accidentally' left somewhere on the way back from a conference. This company does useful stuff. And one new product line being started is big calculators; they have space for advertising and can be useful.

But cost savings is an important issue, and the calculator you are to develop uses a simple (and very cheap) Microchip PIC16 microcontroller. Consider that the platform (i.e. the calculator) will have a 7-segment LCD display (a multiplexer is used to select one of the 12 7-segment characters to control at a time), the number pad (which is connected to serially by UART). Those are the main parts, besides a solar panel and battery. See **Figure 1** block diagram showing main pieces of the calculator on the next page (it's casually called the 'Big Little Calculator' since it is a big size but can do little in the way of calculations, compared to e.g. a scientific calculator). Although it is planned that the production model and the prototyping testing model will have exactly the same PCB, the testing model will have a few more features, allowing the PIC16 to be programmed in place (as opposed to first having the PIC programmed and then inserting it, programmed, into the chip slot). The testing board has an actual reset button, that will be manually soldered on by yourself, or one of the technicians in the assembly plant (they're often happy to do things to help with new products). The advantage of the reset button is quicker code and testing, instead of having to rely on the reset happening from the programmer, or turning off/on the power, the pushbutton can be pressed.

Now, you know what the system being designed is about. And you probably know how it would be used (what the runtime environment would be like). In this case, the execution environment would have much overlap with the runtime environment... but still, the *execution* environment involves a bit more than responding to keypresses. The execution environment may have situations where the lighting is not sufficient to power the solar cells, relying on the battery ... might need a means to switch between the power sources (and what happens then?). Multiple keys might be pressed at once, but the *runtime* environment might not have any facility to detect this, e.g. with a simple serially controlled numpad it might return the code for only one of the keys that is pressed, for instance the keys may be scanned from left to right, top row to bottom row, and soon as a key is detected as down it returns the code for that key and restarts standing (so if you pressed 7 and 3 together, it just returns 7 and there's no indication 3 was pressed too).

YOUR TASKS: Think about and make brief notes on what would be used or would be a part of:

- Run-time System
- Run-time Environment
- Development Environment
- Execution Environment

Sample solutions will be given and discussed in the next lecture.

Big Little Calculator – a simple embedded system example

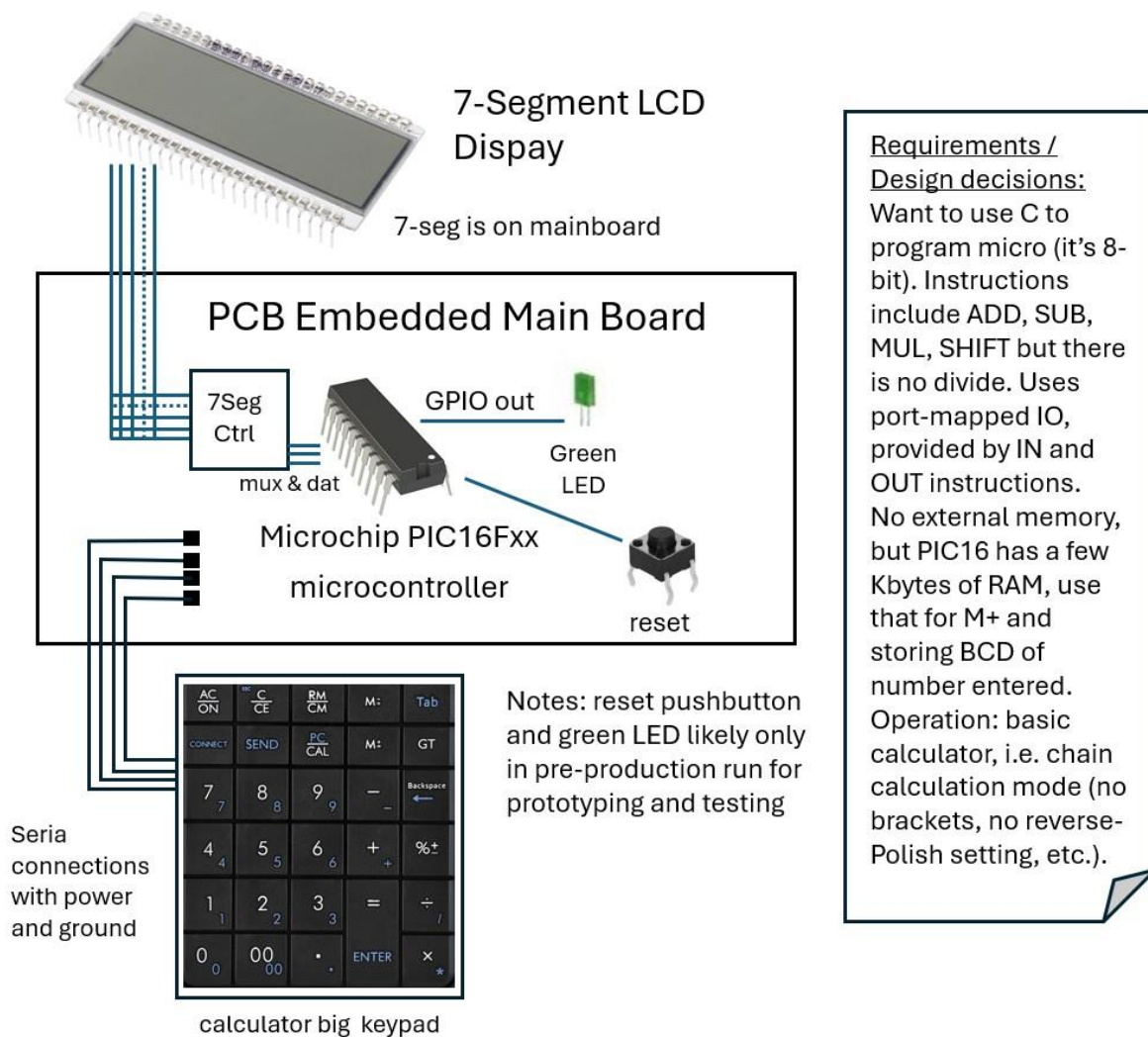


Figure 1: The main parts of Big Little Calculator and some points of requirements and operation.